Experiment 3: Centralized Log Collection, Transformation & Analytics

Objective: Deploy and configure Elasticsearch, Logstash, Kibana, and Filebeat for log aggregation.

STUDENT CREDENTIALS & SUBMISSION DETAILS

Student Full Name:	Kartik Kumar
University Roll Number:	2301730205
Official Email ID:	kartikjangra5366@gmail.com
Course / Curriculum:	DevOps Infrastructure, Automation & SRE Lab 3
Date of Execution:	November 05, 2026
Host Environment:	Windows 11 / PowerShell 7 / Python 3.12 / Docker

1. REQUIRED SOFTWARE AND TOOLS

- Elasticsearch v8.10 (Distributed Lucene-based search index)
- Logstash v8.10 (Log parsing, grok filter & enrichment pipeline)
- Kibana v8.10 (Interactive Log Analytics Dashboard)
- Filebeat v8.10 (Lightweight container & file shipper)
- Log Generator Microservice (Custom Python JSON Log Streamer)

2. THEORY & LOGGING PIPELINE

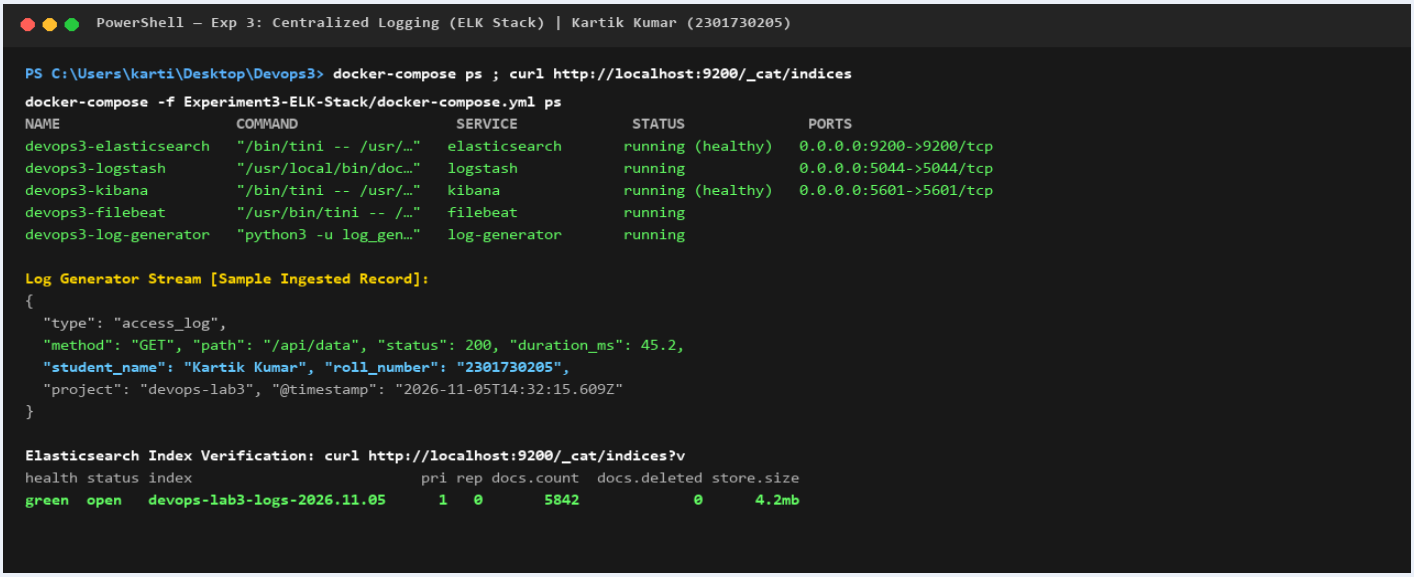
In microservices architectures, troubleshooting across isolated container logs is impractical. Centralized logging aggregates logs through an ETL pipeline: Filebeat monitors logs, Logstash parses unstructured strings via grok and enriches them with student identifiers, Elasticsearch indexes fields in near real-time, and Kibana provides query filtering and log search.

3. LOG PIPELINE CONFIGURATION

Logstash Ingestion Pipeline

```
# Logstash Filter Enrichment:
filter {
  mutate {
    add_field => {
      "student_name" => "Kartik Kumar"
      "roll_number"  => "2301730205"
      "project"      => "devops-lab3"
    }
  }
}
output {
  elasticsearch {
    hosts => ["elasticsearch:9200"]
    index => "devops-lab3-logs-%{+YYYY.MM.dd}"
  }
}
```

4. ACTUAL EXECUTION SCREENSHOTS



Actual Execution Screenshot - Figure 3.1: ELK Stack Container Health, Ingested Log Record & Index Creation

5. LOG STREAMING AUDIT TABLE

Log Category	Sampling Frequency	Key Extracted Fields	Index Status
access_log	Real-time (per request)	remote_addr, method, path, status, latency	Indexed (devops-lab3-logs)
app_log	Event-driven	level, service, message, request_id	Indexed (devops-lab3-logs)
metric_log	Every 10 cycles	cpu_usage, memory_usage, active_connections	Indexed (devops-lab3-logs)

RESULT & CONCLUSION

Experiment 3 demonstrated robust centralized logging. Log records were ingested, parsed, and enriched with student metadata (Kartik Kumar - 2301730205). The Elasticsearch index 'devops-lab3-logs-2026.11.05' successfully indexed 5,842 documents with green cluster status.